



AKGEC SKILLS FOUNDATION

Training Program Details

FAB LAB CENTRE OF EXCELLENCE

Training/Course Title:	Short Term Training Program Embedded / Internet Of Things
Target Audience:	Students – B. Tech / Diploma
Fee:	Rs. 7000
Duration:	40 Hrs.
Module	▪ Introduction to Embedded Systems and IoT ecosystem ▪ Basics of electronics: sensors, actuators, power supply, and interfacing ▪ Microcontroller fundamentals (Arduino / ESP32 / ESP8266 / Raspberry Pi – overview) ▪ Programming fundamentals for embedded systems (C / Arduino IDE) ▪ Digital and analog I/O interfacing ▪ Sensor interfacing (temperature, humidity, motion, gas, light, etc.) ▪ Actuator control (LEDs, relays, motors, buzzers) ▪ Communication protocols: UART, I2C, SPI ▪ Wireless communication fundamentals: Wi-Fi, Bluetooth, MQTT ▪ IoT architecture and cloud platforms (ThingSpeak / Blynk / Firebase – basics) ▪ Data acquisition, monitoring, and visualization ▪ Introduction to PCB design and basic prototyping (breadboard & FABLAB tools) ▪ IoT security fundamentals and best practices ▪ Mini project development using Embedded Systems & IoT ▪ Testing, debugging, and documentation of IoT systems
Outcome:	▪ Understand the fundamentals of embedded systems and IoT architecture ▪ Identify and interface sensors and actuators with microcontrollers ▪ Write, upload, and debug embedded programs independently ▪ Establish wired and wireless communication between devices ▪ Connect embedded systems to cloud platforms for real-time data monitoring ▪ Design and implement a basic IoT-based solution ▪ Develop problem-solving and prototyping skills using FABLAB resources

Certification:	Upon successful completion, participants will be awarded a Globally recognized joint certification from FAB LAB and AKGEC.	
Program Coordinator:	Name: Mr. Himanshu Tripathi	Contact number: 9873812070
Payment Mode:		